

341. For $a \in \mathbb{C}$, let $A = \{z \in \mathbb{C} : \operatorname{Re}(a + \bar{z}) > \operatorname{Im}(\bar{a} + z)\}$ and $B = \{z \in \mathbb{C} : \operatorname{Re}(a + \bar{z}) < \operatorname{Im}(\bar{a} + z)\}$. Then among the two statements :

(S1) : If $\operatorname{Re}(a), \operatorname{Im}(a) > 0$, then the set A contains all the real numbers

(S2) : If $\operatorname{Re}(a), \operatorname{Im}(a) < 0$, then the set B contains all the real numbers

(A) only (S1) is true (B) both are false (C) only (S2) is true (D) both are true

$a \in \mathbb{C}$ के लिए, माना $A = \{z \in \mathbb{C} : \operatorname{Re}(a + \bar{z}) > \operatorname{Im}(\bar{a} + z)\}$ तथा $B = \{z \in \mathbb{C} : \operatorname{Re}(a + \bar{z}) < \operatorname{Im}(\bar{a} + z)\}$ हैं। तो दो कथनों :

(S1) : यदि $\operatorname{Re}(a), \operatorname{Im}(a) > 0$ है, तो सभी वास्तविक संख्याएँ A में हैं

(S2) : यदि $\operatorname{Re}(a), \operatorname{Im}(a) < 0$ है, तो सभी वास्तविक संख्याएँ B में हैं

(A) केवल (S1) सही है (B) दोनों गलत है (C) केवल (S2) सही है (D) दोनों सही है

Ans. B

Sol. Let $a = x_1 + iy_1, z = x + iy$

Now $\operatorname{Re}(a + \bar{z}) > \operatorname{Im}(\bar{a} + z)$

$$\therefore x_1 + x > -y_1 + y$$

$$x_1 = 2, y_1 = 10, x = -12, y = 0$$

Given inequality is not valid for these values.

S1 is false.

Now $\operatorname{Re}(a + \bar{z}) < \operatorname{Im}(\bar{a} + z)$

$$x_1 + x < -y_1 + y$$

$$x_1 = -2, y_1 = -10, x = 12, y = 0$$

Given inequality is not valid for these values.

S2 is false.

342. Let C be the circle in the complex plane with centre $z_0 = \frac{1}{2}(1 + 3i)$ and radius $r = 1$. Let $z_1 = 1 + i$ and the complex

number z_2 be outside the circle C such that $|z_1 - z_0||z_2 - z_0| = 1$. If z_0, z_1 and z_2 are collinear, then the smaller value of $|z_2|^2$ is equal to :

माना सम्मिश्र तल में वृत्त C का केन्द्र $z_0 = \frac{1}{2}(1 + 3i)$ तथा त्रिज्या $r = 1$ हैं। माना $z_1 = 1 + i$ है तथा वृत्त के बाहर सम्मिश्र संख्या z_2

इस प्रकार है कि $|z_1 - z_0||z_2 - z_0| = 1$ है। यदि z_0, z_1 तथा z_2 संरेख हैं, तो $|z_2|^2$ का छोटा मान बराबर है :

(A) $\frac{7}{2}$ (B) $\frac{13}{2}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Ans. D

Sol. $|z_1 - z_0| = \left| \frac{1-i}{2} \right| = \frac{1}{\sqrt{2}}$

$$\Rightarrow |z_2 - z_0| = \sqrt{2}; \text{ centre } \left(\frac{1}{2}, \frac{3}{2}\right)$$

$$z_0 \left(\frac{1}{2}, \frac{3}{2}\right) \text{ and } z_1(1,1)$$

$$\tan \theta = -1 \Rightarrow \theta = 135^\circ$$

$$z_2 \left(\frac{1}{2} + \sqrt{2} \cos 135^\circ, \frac{3}{2} + \sqrt{2} \sin 135^\circ\right)$$

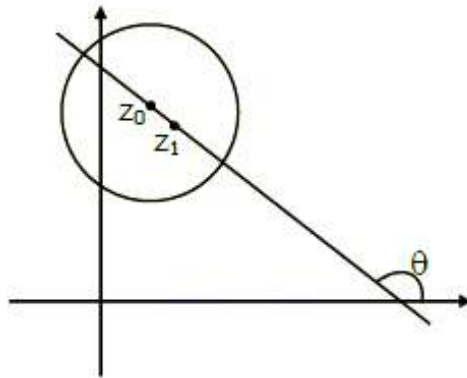
or

$$\left(\frac{1}{2} - \sqrt{2} \cos 135^\circ, \frac{3}{2} - \sqrt{2} \sin 135^\circ\right)$$

$$\Rightarrow z_2 \left(-\frac{1}{2}, \frac{5}{2}\right) \text{ or } z_2 \left(\frac{3}{2}, \frac{1}{2}\right)$$

$$\Rightarrow |z_2|^2 = \frac{26}{4}, \frac{5}{2}$$

$$\Rightarrow |z_2|_{\min} = \frac{5}{2}$$



343. Let α, β be the roots of the equation $x^2 - \sqrt{2}x + 2 = 0$. Then $\alpha^{14} + \beta^{14}$ is equal to :

माना समीकरण $x^2 - \sqrt{2}x + 2 = 0$ के मूल α, β हैं। तो $\alpha^{14} + \beta^{14}$ बराबर है :

- (A) -64 (B) -128 (C) $-64\sqrt{2}$ (D) $-128\sqrt{2}$

Ans. B

Sol. $x^2 - \sqrt{2}x + 2 = 0$

$$x = \frac{\sqrt{2} \pm \sqrt{2-8}}{2} = \frac{\sqrt{2} \pm \sqrt{6}i}{2}$$

$$\alpha = \frac{\sqrt{2} + \sqrt{6}i}{2} = \sqrt{2}e^{\frac{i\pi}{3}} \text{ \& } \beta = \sqrt{2}e^{\frac{-i\pi}{3}}$$

$$\alpha^{14} = 2^7 e^{\frac{i14\pi}{3}} = 128 \left[e^{\frac{i2\pi}{3}} \right]$$

$$\beta^{14} = 128 \left[e^{\frac{-i2\pi}{3}} \right]$$

$$\alpha^{14} + \beta^{14} = 128(2) \cos\left(\frac{2\pi}{3}\right) = -128$$

344. Let $S = \{z \in \mathbb{C} : \bar{z} = i(z^2 + \operatorname{Re}(\bar{z}))\}$. The $\sum_{z \in S} |z|^2$ is equal to :

माना $S = \{z \in \mathbb{C} : \bar{z} = i(z^2 + \operatorname{Re}(\bar{z}))\}$ है। तो $\sum_{z \in S} |z|^2$ बराबर है :

(A) $\frac{5}{2}$

(B) 4

(C) 3

(D) $\frac{7}{2}$

Ans. B

Sol. Let $z = x + iy$

$$\bar{z} = i(z^2 + \operatorname{Re}(\bar{z}))$$

$$\Rightarrow x - iy = i(x^2 - y^2 + 2ixy + x)$$

$$x - iy = i(x^2 - y^2 + x) - 2xy$$

$$x = -2xy \Rightarrow x(2y + 1) = 0$$

$$\Rightarrow x = 0, y = \frac{-1}{2} \quad \dots(i)$$

$$-y = x^2 - y^2 + x \quad \dots(ii)$$

Case (1) $x = 0$

$$(ii) \Rightarrow -y = -y^2 \Rightarrow y^2 - y = 0 \Rightarrow y = 0, 1$$

$$z = 0, i$$

$$\text{Case (2) } y = \frac{-1}{2}$$

$$(ii) \Rightarrow \frac{1}{2} = x^2 - \frac{1}{4} + x \Rightarrow x^2 + x - \frac{3}{4} = 0$$

$$4x^2 + 4x - 3 = 0 \Rightarrow (2x - 1)(2x + 3) = 0$$

$$x = \frac{1}{2}, \frac{-3}{2}$$

$$z = \frac{1}{2} - \frac{1}{2}i, \frac{-3}{2} - \frac{1}{2}i$$

$$\sum |z|^2 = 0 + 1 + \frac{1}{2} + \frac{5}{2} = 4$$

345. $A = \{z \in \mathbb{C} : |z - 2 - i| = 3\}$, $B = \{z \in \mathbb{C} : \operatorname{Re}(z - iz) = 2\}$ and $S = A \cap B$. Then $\sum_{z \in S} |z|^2$ is equal to _____.

माना $A = \{z \in \mathbb{C} : |z - 2 - i| = 3\}$, $B = \{z \in \mathbb{C} : \operatorname{Re}(z - iz) = 2\}$ और $S = A \cap B$ हैं। तो $\sum_{z \in S} |z|^2$ बराबर है _____.

Ans. 22

Sol. $A = \{z \in \mathbb{C} : |z - 2 - i| = 3\}$

$$B = \{z \in \mathbb{C} : \operatorname{Re}(z - iz) = 2\}$$

$$S = A \cap B$$

$$\text{then } \sum_{z \in S} |z|^2 = ??$$

$$\text{Sol. } A : (x-2)^2 + (y-1)^2 = 9 \quad \dots(i)$$

$$B : \operatorname{Re} \{(x+y) + i(y-x)\} = 2$$

$$x+y=2 \quad \dots(ii)$$

Solving (i) & (ii)

$$x = \frac{3 \pm \sqrt{17}}{2}; y = \frac{1 \mp \sqrt{17}}{2}$$

$$\sum_{z \in S} |z|^2 = \frac{1}{4} [2 \times 26 + 2 \times 18]$$

$$\Rightarrow \frac{88}{4}$$

$$\Rightarrow 22$$

346. If α is a root of the equation $x^2 + x + 1 = 0$ and $\sum_{k=1}^n \left(\alpha^k + \frac{1}{\alpha^k} \right)^2 = 20$, then n is equal to _____.

यदि समीकरण $x^2 + x + 1 = 0$ का एक मूल α है और $\sum_{k=1}^n \left(\alpha^k + \frac{1}{\alpha^k} \right)^2 = 20$ है, तो n बराबर है _____.

Ans. 11

Sol. $\alpha = w, w^2$

$$\sum_{k=1}^n \left(w^k + \frac{1}{w^k} \right)^2 = 20$$

$$\left(w + \frac{1}{w} \right)^2 + \left(w^2 + \frac{1}{w^2} \right)^2 + \left(w^3 + \frac{1}{w^3} \right)^2 + \left(w^4 + \frac{1}{w^4} \right)^2 + \dots = 20$$

$$\Rightarrow (w + w^2)^2 + (w^2 + w)^2 + 2^2 + (w + w^2)^2 + \dots = 20$$

$$\Rightarrow (-1)^2 + (-1)^2 + 4 + (-1)^2 + \dots = 20$$

$$\Rightarrow n = 11$$

347. If $\alpha = 1 + \sum_{r=1}^6 (-3)^{r-1} {}^{12}C_{2r-1}$, then the distance of the point $(12, \sqrt{3})$ from the line $\alpha x - \sqrt{3}y + 1 = 0$ is _____.

यदि $\alpha = 1 + \sum_{r=1}^6 (-3)^{r-1} {}^{12}C_{2r-1}$ है, तो बिन्दु $(12, \sqrt{3})$ की रेखा $\alpha x - \sqrt{3}y + 1 = 0$ से दूरी है _____.

Ans. 5

Sol. $\alpha = 1 + \sum_{r=1}^6 (-1)^{r-1} {}^{12}C_{2r-1} 3^{r-1}$

$$\alpha = 1 + \sum_{r=1}^6 {}^{12}C_{2r-1} \frac{(\sqrt{3}i)^{2r-1}}{\sqrt{3}i} \quad i = \text{iota, let } \sqrt{3}i = x$$

$$\alpha = 1 + \frac{1}{\sqrt{3}i} ({}^{12}C_1 x + {}^{12}C_3 x^3 + \dots + {}^{12}C_{11} x^{11})$$

$$= 1 + \frac{1}{\sqrt{3}i} \left(\frac{(1 + \sqrt{3}i)^{12} - (1 - \sqrt{3}i)^{12}}{2} \right)$$

$$= 1 + \frac{1}{\sqrt{3}i} \left(\frac{(-2w^2)^{12} - (2w)^{12}}{2} \right) = 1$$

So, distance of $(12, \sqrt{3})$ from $x - \sqrt{3}y + 1 = 0$ is

$$\frac{12 - 3 + 1}{2} = 5$$

348. Let integers $a, b \in [-3, 3]$ be such that $a + b \neq 0$. Then the number of all possible ordered pairs (a, b) , for which

$$\left| \frac{z-a}{z+b} \right| = 1 \text{ and } \begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 1, z \in \mathbb{C}, \text{ where } \omega \text{ and } \omega^2 \text{ are the roots of } x^2 + x + 1 = 0, \text{ is equal to}$$

_____.

माना कि पूर्णांक $a, b \in [-3, 3]$ इस प्रकार है कि $a + b \neq 0$ है। तब उन सभी संभव क्रमित युग्मों (a, b) , जिनके लिए $\left| \frac{z-a}{z+b} \right| = 1$ और

$$\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 1, z \in \mathbb{C} \text{ हैं, जहाँ } \omega \text{ और } \omega^2 \text{ समीकरण } x^2 + x + 1 = 0 \text{ के मूल हैं, की संख्या बराबर बराबर है } \underline{\hspace{2cm}}.$$

Ans. 10

Sol. $a, b \in \mathbb{I}, -3 \leq a, b \leq 3, a + b \neq 0$

$$|z-a| = |z+b|$$

$$\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 1$$

$$\Rightarrow \begin{vmatrix} z & z & z \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 1$$

$$\Rightarrow z \begin{vmatrix} 1 & 1 & 1 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 1$$

$$\Rightarrow z \begin{vmatrix} 1 & 0 & 0 \\ \omega & z+\omega^2-\omega & 1-\omega \\ \omega^2 & 1-\omega^2 & z+\omega-\omega^2 \end{vmatrix} = 1$$

$$\Rightarrow z^3 = 1$$

$$\Rightarrow z = \omega, \omega^2, 1$$

$$\text{Now } |1-a| = |1+b|$$

$$\Rightarrow 10 \text{ pairs}$$

349. Let x_1, x_2, x_3, x_4 be the solution of the equation $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$ and

$$(4+x_1^2)(4+x_2^2)(4+x_3^2)(4+x_4^2) = \frac{125}{16}m. \text{ Then the value of } m \text{ is } \underline{\hspace{2cm}}.$$

माना x_1, x_2, x_3, x_4 समीकरण $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$ के हल हैं तथा

$$(4+x_1^2)(4+x_2^2)(4+x_3^2)(4+x_4^2) = \frac{125}{16}m \text{ है। तो } m \text{ का मान है } \underline{\hspace{2cm}}$$

Ans. 221

Sol. $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$

x_1, x_2, x_3, x_4 are the solution of equation

$$4x^4 + 8x^3 - 17x^2 - 12x + 9 = 4(x-x_1)(x-x_2)(x-x_3)(x-x_4)$$

put $x = 2i$

$$4(2i)^4 + 8(2i)^3 - 17(2i)^2 - 12(2i) + 9$$

$$= 4[(2i-x_1)(2i-x_2)(2i-x_3)(2i-x_4)]$$

$$\Rightarrow 64 - 64i + 68 - 24i + 9$$

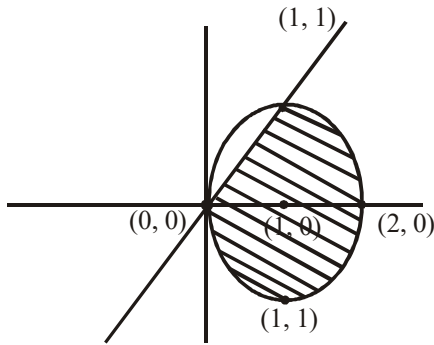
$$\begin{aligned} \Rightarrow |141 - 88i| &= 4|(2i - x_1)(2i - x_2)(2i - x_3)(2i - x_4)| \\ \Rightarrow (141)^2 + (88)^2 &= 16[(4 + x_1^2)(4 + x_2^2)(4 + x_3^2)(4 + x_4^2)] \\ \Rightarrow \frac{19881 + 7744}{16} &= (4 + x_1^2)(4 + x_2^2)(4 + x_4^2) \\ \Rightarrow \frac{27625}{16} &= \frac{125}{16}m \\ m &= 221 \end{aligned}$$

350. The sum of the square of the modulus of the elements in the set $\{z = a + ib : a, b \in \mathbb{Z}, z \in \mathbb{C}, |z - 1| \leq 1, |z - 5| \leq |z - 5i|\}$ is _____.

समुच्चय $\{z = a + ib : a, b \in \mathbb{Z}, z \in \mathbb{C}, |z - 1| \leq 1, |z - 5| \leq |z - 5i|\}$ के अवयवों के मापांक के वर्गों का योग है _____.

Ans. 9

Sol.



$$\begin{aligned} |Z - 1| &\leq 1 \\ |Z - 5| &\leq |Z - 5i| \\ (a, b) &= \{(0, 0), (1, 1), (1, 0), (1, -1), (2, 0)\} \\ |Z|^2 &= \{0, 2, 1, 2, 4\} \\ \text{Sum} &= 9 \end{aligned}$$

351. If α satisfies the equation $x^2 + x + 1 = 0$ and $(1 + \alpha)^7 = A + B\alpha + C\alpha^2$, $A, B, C \geq 0$, then $5(3A - 2B - C)$ is equal to _____.

यदि α समीकरण $x^2 + x + 1 = 0$ को संतुष्ट करता है तथा $(1 + \alpha)^7 = A + B\alpha + C\alpha^2$, $A, B, C \geq 0$ हैं, तो $5(3A - 2B - C)$ बराबर है _____।

Ans. 5

Sol. $\alpha^2 + \alpha + 1 = 0 \left\langle \frac{\omega}{\omega^2} \right\rangle \omega^3 = 1 (x^2 + x + 1 = 0) 0 \left\langle \frac{\omega}{\omega^2} \right\rangle \omega^3 = 1 \text{ let } \alpha = \omega$

$$(1 + \alpha)^7 = 1^7 + {}^7C_1 \alpha + {}^7C_2 \alpha^2 + {}^7C_3 \alpha^3 + {}^7C_4 \alpha^4 + {}^7C_5 \alpha^5 + {}^7C_6 \alpha^6 + {}^7C_7 \alpha^7$$

$$= 1 + 7\alpha + 21\alpha^2 + 35\alpha^3 + 35\alpha^4 + 21\alpha^5 + 7\alpha^6 + \alpha^7$$

$$\because \alpha = \omega$$

$$= 1 + 7\omega + 21\omega^2 + 35\omega^3 + 35\omega^4 + 21\omega^5 + 7\omega^6 + \omega^7$$

$$(\because \omega^3 = 1)$$

$$= 1 + 7\omega + 21\omega^2 + 35 + 35\omega + 21\omega + 7 + \omega$$

$$= 43 + 43\omega + 42\omega^2$$

$$= 43 + 43\alpha + 42\alpha^2 = A + B\alpha + C\alpha^2$$

$$\Rightarrow A = 43 \mid B = 43 \mid C = 42$$

$$5(129 - 86 - 42)$$

$$= 5(129 - 128) = 5$$

352. Let the complex numbers α and $\frac{1}{\alpha}$ lie on the circle $|z - z_0|^2 = 4$ and $|z - z_0|^2 = 16$ respectively, where $z_0 = 1 + i$. Then, the value of $100 |\alpha|^2$ is _____.

माना सम्मिश्र संख्याएँ α तथा $\frac{1}{\alpha}$ क्रमशः वृत्तों $|z - z_0|^2 = 4$ तथा $|z - z_0|^2 = 16$ पर हैं, जहाँ

$z_0 = 1 + i$ है। तो $100 |\alpha|^2$ का मान है _____।

Ans. 20

Sol. $|\alpha - z_0|^2 = 4$

$$\Rightarrow (\alpha - z_0)(\bar{\alpha} - \bar{z}_0) = 4$$

$$|\alpha|^2 - \alpha \bar{z}_0 - \bar{\alpha} z_0 + |z_0|^2 = 4$$

$$|\alpha|^2 - \alpha \bar{z}_0 - \bar{\alpha} z_0 = 2 \quad \text{_____ (i)}$$

$$(ii) \left| \frac{1}{\alpha} - z_0 \right|^2 = 16 \Rightarrow (1 - \bar{\alpha} z_0)(1 - \alpha \bar{z}_0) = 16 |\alpha|^2$$

$$\Rightarrow 1 - \alpha \bar{z}_0 - \bar{\alpha} z_0 + |\alpha|^2 \cdot 2 = 16 |\alpha|^2 \quad \text{_____ (ii)}$$

$$\text{from (i) and (ii)} \quad -1 - |\alpha|^2 = 2 - 16 |\alpha|^2 \Rightarrow 15 |\alpha|^2 = 3 \Rightarrow 5 |\alpha|^2 = 1$$

$$100 |\alpha|^2 = 20$$

353. Let α, β the roots of the equation $x^2 - x + 2 = 0$ with $\text{Im}(\alpha) > \text{Im}(\beta)$. Then $\alpha^6 + \alpha^4 + \beta^4 - 5\alpha^2$ is equal to _____.

माना समीकरण $x^2 - x + 2 = 0$ के मूल α, β हैं तथा $\text{Im}(\alpha) > \text{Im}(\beta)$ है। तो $\alpha^6 + \alpha^4 + \beta^4 - 5\alpha^2$ बराबर है _____.

Ans. 13

Sol. $x^2 + 2 = x$

$$\Rightarrow x^4 + 4 + 4x^2 = x^2$$

So,

$$x^4 = -4 - 3x^2$$

$$\Rightarrow x^6 = -4x^2 - 3x^4$$

$$\Rightarrow x^6 = 12 + 5x^2$$

Now,

$$\alpha^6 - 5\alpha^2 = 12$$

$$\text{So } \alpha^4 + \beta^4 + 12$$

$$= 4 - 3(\alpha^2 + \beta^2)$$

$$= 4 - 3[1 - 4]$$

$$= 13$$

354. Let α, β be the roots of the equation $x^2 - \sqrt{6}x + 3 = 0$ such that $\text{Im}(\alpha) > \text{Im}(\beta)$. Let a, b be integers not divisible by 3 and n be a natural number such that $\frac{\alpha^{99}}{\beta} + \alpha^{98} = 3^n(a + ib)$, $i = \sqrt{-1}$. Then $n + a + b$ is equal to _____.

माना समीकरण $x^2 - \sqrt{6}x + 3 = 0$ के मूल α, β हैं तथा $\text{Im}(\alpha) > \text{Im}(\beta)$ है। माना 3 से अभाज्य पूर्णांक a, b हैं तथा घन पूर्णांक

n के लिए $\frac{\alpha^{99}}{\beta} + \alpha^{98} = 3^n(a + ib)$, $i = \sqrt{-1}$ है। तो $n + a + b$ बराबर है _____

Ans. 49

Sol. $x^2 - \sqrt{6}x + 3 = 0$ $\begin{matrix} \nearrow \alpha \\ \searrow \beta \end{matrix}$

$$x = \frac{\sqrt{6} \pm i\sqrt{6}}{2} = \frac{\sqrt{6}}{2}(1 \pm i)$$

$$\alpha = \sqrt{3} \left(e^{i\frac{\pi}{4}} \right), \beta = \sqrt{3} \left(e^{-i\frac{\pi}{4}} \right)$$

$$\therefore \frac{\alpha^{99}}{\beta} + \alpha^{98} = \alpha^{98} \left(\frac{\alpha}{\beta} + 1 \right)$$

$$= \frac{\alpha^{98}(\alpha + \beta)}{\beta} = 3^{49} \left(e^{i99\frac{\pi}{4}} \right) \times \sqrt{2}$$

$$= 3^{49}(-1 + i)$$

$$= 3^n(a + ib)$$

$$\therefore n = 49, a = -1, b = 1$$

$$\therefore n + a + b = 49 - 1 + 1 = 49$$

355. If α denotes the number of solutions of $|1 - i|^x = 2^x$ and $\beta = \left(\frac{|z|}{\arg(z)} \right)$, where $z = \frac{\pi}{4}(1+i)^4 \left[\frac{1-\sqrt{\pi}i}{\sqrt{\pi}+i} + \frac{\sqrt{\pi}-i}{1+\sqrt{\pi}i} \right]$, $i = \sqrt{-1}$, then the distance of the point (α, β) from the line $4x - 3y = 7$ is _____.

यदि $|1 - i|^x = 2^x$ के हलों की संख्या α है तथा $\beta = \left(\frac{|z|}{\arg(z)} \right)$ है, जहाँ $z = \frac{\pi}{4}(1+i)^4 \left[\frac{1-\sqrt{\pi}i}{\sqrt{\pi}+i} + \frac{\sqrt{\pi}-i}{1+\sqrt{\pi}i} \right]$, $i = \sqrt{-1}$ है, तो रेखा $4x - 3y = 7$ से बिन्दु (α, β) की दूरी है _____.

Ans. 3

Sol.

$$z = \frac{\pi}{4}(1+i)^4 \left[\frac{1+\pi+\pi+1}{i(1+\pi)-\sqrt{\pi}+\sqrt{\pi}} \right]$$

$$= \frac{\pi}{4}(1+i)^4 \frac{2}{i} = \frac{\pi}{2} \frac{(1+i)^4}{i} = \frac{-\pi}{2} i(1+i)^4$$

$$|z| = \left| \frac{-\pi}{2} \right| |i| |1+i|^4 = 2\pi$$

$$z = \frac{-\pi}{2}(-4i) = 2\pi i \Rightarrow \arg(z) = \frac{\pi}{2}$$

$$\Rightarrow \beta = \frac{2\pi}{\frac{\pi}{2}} = 4$$

Also $|1 - i|^x = 2^x$

$$(\sqrt{2})^x = 2^x$$

$$\Rightarrow x = 0$$

$$\Rightarrow 1 \text{ solution} \Rightarrow \alpha = 1$$

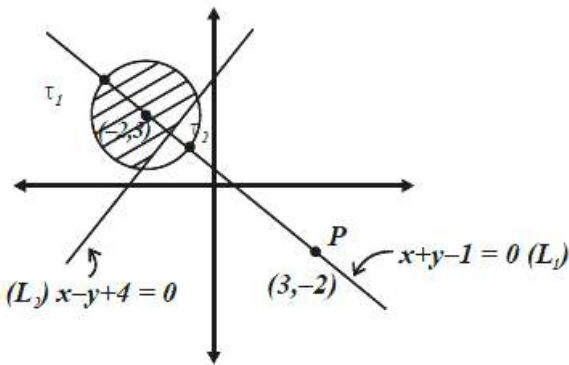
Now perpendicular distance of $4x - 3y = 7$ from $(1, 4)$

$$\left| \frac{4(1) - 3(4) - 7}{5} \right| = 3$$

- 356.** Let $P = \{z \in \mathbb{C} : |z + 2 - 3i| \leq 1\}$ and $Q = \{z \in \mathbb{C} : z(1+i) + \bar{z}(1-i) \leq -8\}$. Let in $P \cap Q$, $|z - 3 + 2i|$ be maximum and minimum at z_1 and z_2 respectively. If $|z_1|^2 + 2|z_2|^2 = \alpha + \beta\sqrt{2}$, where α, β are integers, then $\alpha + \beta$ equals _____.

माना $P = \{z \in \mathbb{C} : |z + 2 - 3i| \leq 1\}$ तथा $Q = \{z \in \mathbb{C} : z(1+i) + \bar{z}(1-i) \leq -8\}$ हैं। माना $P \cap Q$ में, $|z - 3 + 2i|$, z_1 तथा z_2 पर क्रमशः अधिकतम तथा निम्नतम है। यदि $|z_1|^2 + 2|z_2|^2 = \alpha + \beta\sqrt{2}$ है, जहाँ α, β पूर्णांक हैं, तो $\alpha + \beta$ बराबर है _____

Ans. 36



Sol.

Clearly for the shaded region z_1 is the intersection of the circle and the line passing through P (L_1) and z_2 is intersection of line L_1 & L_2

$$\text{Circle : } (x + 2)^2 + (y - 3)^2 = 1$$

$$L_1 : x + y - 1 = 0$$

$$L_2 : x - y + 4 = 0$$

On solving circle & L_1 we get

$$z_1 : \left(-2 - \frac{1}{\sqrt{2}}, 3 + \frac{1}{\sqrt{2}} \right)$$

On solving L_1 and L_2 is intersection of line L_1 & L_2

$$\text{we get } z_2 : \left(\frac{-3}{2}, \frac{5}{2} \right)$$

$$|z_1|^2 + 2|z_2|^2 = 14 + 5\sqrt{2} + 17$$

$$= 31 + 5\sqrt{2}$$

$$\text{So, } \alpha = 31$$

$$\beta = 5$$

$$\alpha + \beta = 36$$

Statistics

357. If the mean and the variance of 6, 4, a, 8, b, 12, 10, 13 are 9 and 9.25 respectively, then $a + b + ab$ is equal to:

यदि 6, 4, a, 8, b, 12, 10, 13 के माध्य और प्रसरण क्रमशः 9 और 9.25 हैं, तो $a + b + ab$ बराबर है :

(A) 100

(B) 105

(C) 106

(D) 103

Ans. D

Sol. Mean = $\frac{6+4+a+8+b+12+10+13}{8} = 9$

$$53 + a + b = 72$$

$$\Rightarrow a + b = 19 \text{ and}$$

And

$$\frac{36+16+a^2+64+b^2+144+100+169}{8} - 9^2 = 9.25$$

$$\Rightarrow \frac{529+a^2+b^2}{8} = 90.25$$

$$\Rightarrow a^2 + b^2 = 193$$

$$\therefore a = 7, b = 12$$

$$\text{Hence, } a + b + ab = 19 + 84 = 103$$

358. Let the Mean and Variance of five observations $x_1 = 1, x_2 = 3, x_3 = a, x_4 = 7$ and $x_5 = b, a > b$, be 5 and 10 respectively. Then the Variance of the observations $n + x_n, n = 1, 2, \dots, 5$ is :

माना पाँच प्रेक्षणों $x_1 = 1, x_2 = 3, x_3 = a, x_4 = 7$ और $x_5 = b, a > b$, के माध्य और प्रसरण क्रमशः 5 और 10 हैं। तो प्रेक्षणों $n + x_n, n = 1, 2, \dots, 5$ का प्रसरण है :

(A) 16

(B) 17.4

(C) 17

(D) 16.4

Ans. A

Sol. $\bar{x} = \frac{\sum x_i}{n} = \frac{1+3+a+7+b}{5} = 5$

$$a + b = 14$$

$$\sigma^2 = \frac{\sum x_i^2}{n} - (\bar{x})^2$$

$$\Rightarrow \frac{1^2+3^2+a^2+7^2+b^2}{5} - 25 = 10$$

$$a^2 + b^2 = 116$$

$$a > b$$

$$a = 10$$

$$b = 4$$

$$n + x_n : 2, 5, 13, 11, 9$$

$$\sigma^2 = \frac{2^2 + 5^2 + 13^2 + 11^2 + 9^2}{5} - \left(\frac{2 + 5 + 13 + 11 + 9}{5} \right)^2$$

$$= 80 - 64 = 16$$

- 359.** Let the mean and the standard deviation of the observation 2, 3, 3, 4, 5, 7, a, b be 4 and $\sqrt{2}$ respectively. Then the mean deviation about the mode of these observations is :

माना प्रेक्षणों 2, 3, 3, 4, 5, 7, a, b के माध्य और मानक विचलन क्रमशः 4 और $\sqrt{2}$ हैं। तो इन प्रेक्षणों का बहुलक के सापेक्ष माध्य विचलन है :

(A) $\frac{3}{4}$

(B) $\frac{1}{2}$

(C) 1

(D) 2

Ans. C

Sol. Mean = $\frac{24 + a + b}{8} = 4$

$$\Rightarrow a + b = 8 \quad \dots(1)$$

$$\text{Variance} = \frac{\sum x_i^2}{n} - (\bar{x})^2 = 2$$

$$\frac{112 + a^2 + b^2}{8} - 16 = 2$$

$$\Rightarrow a^2 + b^2 = 32 \quad \dots(2)$$

$$\text{from (1) \& (ii)} \Rightarrow a = 4, b = 4$$

$$\text{Mode} = 4$$

Mean deviation about mode

$$\Rightarrow \frac{2 + 1 + 1 + 0 + 1 + 3 + 0 + 0}{8} = 1$$

- 360.** The mean and standard deviation of 100 observations are 40 and 5.1, respectively. By mistake one observation is taken as 50 instead of 40. If the correct mean and the correct standard deviation are μ and σ respectively, then $10(\mu + \sigma)$ is equal to :

100 प्रेक्षणों का माध्य और मानक विचलन क्रमशः 40 और 5.1 हैं। त्रुटि से एक प्रेक्षण को 40 के स्थान पर 50 लिया गया है। यदि सही माध्य और सही मानक विचलन क्रमशः μ और σ हैं, तो $10(\mu + \sigma)$ बराबर है :

(A) 451

(B) 447

(C) 449

(D) 445

Ans. C

Sol. $\mu = \frac{100 \times 40 - 50 + 40}{100} = 39.9$

$$(5.1)^2 = \frac{\sum x_i^2}{100} - (\bar{x})^2$$

$$\begin{aligned}\sum x_i^2 &= 100 \times 40^2 + 100 \times (5.1)^2 \\ &= 162601\end{aligned}$$

$$\sigma^2 = \frac{\sum x_i^2 - 52^2 + 40^2}{100} - \mu^2 = 25$$

$$100 (\mu + \sigma) = 10(39.9 + 5) = 449$$